1000-class Face Classification: Experiment Report

1 What the task was

1000-class face classification on 64×64 RGB images. Train 93,134 (24–221 per class, avg. 93), val 2,000 (2 per class), test 2,394. Metric is accuracy. I ended at 86.40% val with a four-model weighted TTA ensemble.

2 What I tried, in order

(See Table 1 at the end of the document.)

3 Things that broke and how I fixed them

3.1 LR restart on a converged model didn't help

Resuming and restarting cosine from 0.05 dropped val below 60% and it never came back. High LR kicked the converged model out of its basin, and 20 epochs wasn't enough to climb back. Lesson: for a converged model, continue from a small LR ($10^{-4} - 10^{-3}$), don't restart the schedule.

3.2 ArcFace wouldn't converge

- Pure ArcFace ($m = 0.5$, $s = 64$) from scratch: loss went from 42 to 54, accuracy stuck at 0.1%.
- Gradient direction was fine; the margin was just too big, over-penalizing the correct class early.
- Tried dynamic margin warm-up, $m = 0.3/0.2$, and pretrained backbone + low-LR fine-tuning. Best was 78.2%, still below the CE + distillation baseline.
- ArcFace is built for million-scale data; it doesn't fit 1000 classes $\times$ 93 samples.

3.3 Small bugs

- DataLoader returned a triple (with indices); evaluate failed to unpack $\rightarrow$ used `batch[0]`, `batch[1]`.
- `torch.load` defaults to `weights_only=True` in PyTorch 2.6+, so loading failed $\rightarrow$ set `weights_only=False`.
- A seed edit to the optimizer args deleted a parenthesis $\rightarrow$ fixed by hand.
- Ensemble script's lambda referenced an undefined variable when sorting by `test_list` $\rightarrow$ index via `TestSet.names`.

4 What worked and what didn't

4.1 Worked (roughly in order of contribution)

1. self-distillation (83.5%, biggest single jump, +4);
2. ensembling (+0.3–0.5 per "strong and different" member);
3. TTA (+0.3);
4. LR restart (helped early, +1.5);
5. blur augmentation (big robustness gain on blurred test data);
6. optimizer diversity (SGD/Adam ensemble, +0.35);
7. mutual distillation (as a diversity source, +0.15).

4.2 Didn't work

1. continued training / LR restart on converged models;
2. bagging (weakens single models, and its diversity overlaps with TTA);
3. ArcFace (data scale mismatch);
4. larger resolution ($64 \rightarrow 128$ only +0.4, saturated).

4.3 What I took away from it

- The bottleneck is data (avg. 93 per class), not model capacity, resolution, or how separable the classes are.
- Ensemble error $\propto$ single-model error $\times$ inter-member error correlation; "strong and different" members are the ones that pay off.
- Change one variable at a time and log val — that's what kept this efficient.

Table 1: Experiment log

#	Method	Val	Worked?	Notes
1	baseline ResNet (3M, SGD)	77.8%	baseline	starting point
2	LR restart (resume + cosine restart)	79.25%	yes	second annealing stage escaped a local optimum
3	blur augmentation	77.0%	partly	val slightly down, but +60 on blurred images; didn't know how blurry test really is
4	self-distillation stage 1 (teacher = 2-model ensemble soft labels, student 11.7M)	83.5%	yes	soft labels carry inter-class similarity
5	stage-2 distillation + 128px upsampling	83.9%	slightly	resolution gain basically saturated
6	ensemble dist64 + dist128 (weight search)	85.55%	yes	different resolutions decorrelate errors
7	TTA (horizontal flip)	85.90%	yes	removes left/right bias
8	keep training 20–60 epochs with LR restart	no gain	no	high LR knocked the converged model out of its basin
9	bagging (60% sample $\times 3$)	77.8–79.2%	weak alone	40% less data hurts each model
10	bagging added to ensemble	85.75%	slightly	5% weight, third viewpoint
11	bagging + TTA	85.75%	no gain	diversity overlaps with TTA
12	Adam version, standalone	82.0%	weaker than SGD	converges faster, generalizes a bit worse
13	Adam added to ensemble + TTA	86.25%	yes	optimizer diversity is real diversity
14	mutual distillation (SGD + Adam soft labels, new student)	82.65%	between teachers	mediocre alone but different error pattern
15	four-model weighted TTA ensemble	86.40%	final	mutual 10 + adam 20 + d64 25 + d128 45
16	ArcFace from scratch	stuck at 0.1%	no	$m = 0.5$ too aggressive for 1000 classes with so few samples
17	ArcFace fine-tune from pretrained backbone	78.2%	below baseline	bottleneck is sample count, not class confusion
18	ArcFace in ensemble	no gain	no	cosine vs. CE logits scale differently, and correlated with dist64